Analysis of Intensity Discrimination Thresholds in Human Auditory Perception

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***Abstract*—This Stage-2 report presents a 2-level full factorial experimental design, analysis, and preliminary predictive model for the intensity Difference Limen (DL) in human auditory perception under the influence of digital bitrate and stimulus parameters. The report summarizes the Stage-1 findings, states the chosen factorial design (factors, levels, replications), presents raw and aggregated results, computes main and interaction effects with confidence intervals, performs ANOVA and regression modeling in coded variables, and shows model adequacy diagnostics. Recommendations for model improvement and next steps are provided.**

I. Recap from Stage 1

Stage 1 of the project focused on developing and validating an experimental framework for measuring intensity discrimination thresholds in human auditory perception. The primary objective was to design a reliable psychophysical experiment capable of estimating the intensity Difference Limen (DL) under the combined influence of physiological hearing mechanisms and digital audio system parameters. This stage involved reviewing relevant psychoacoustic literature, defining the experimental problem, developing a web-based testing platform, and conducting pilot experiments to verify the experimental methodology.

A. Key outcomes from Stage 1

- **Main conclusions:** The pilot experiments confirmed that the experimental setup could reliably measure intensity Difference Limens using an adaptive psychophysical procedure. The results revealed two clear trends consistent with established psychoacoustic theory. First, the measured DL decreased as the pedestal intensity increased, demonstrating the classical "near-miss" to Weber's Law, where discrimination improves more rapidly than predicted by a constant Weber fraction. Second, the results showed that audio compression significantly affects perceptual sensitivity: reducing the bitrate from 128 kbps to 32 kbps increased the measured DL by approximately 0.7–0.9 dB, indicating that compression artefacts degrade the listener's ability to detect small intensity changes.
- **Learnings about the system:** A web-based auditory testing platform was successfully implemented to conduct the experiments remotely. The experiment employed a two-alternative forced-choice (2AFC) paradigm combined with a three-down/one-up adaptive staircase procedure that converges to the 79.4% correct point on the psychometric function. Participants were presented with two sequential tone intervals and asked to identify the louder stimulus. The adaptive staircase dynamically adjusted the intensity difference between the two tones based on participant responses. Because the experiment was conducted remotely using participant-provided headphones, absolute sound pressure levels could not be controlled. Therefore, all intensity measurements were defined relative to a participant-selected comfortable listening level.
- **Sources of variability:** Several sources of experimental variability were identified during the pilot study. Environmental factors such as background noise in the participant's surroundings could mask low-level stimuli and affect threshold measurements. Hardware differences in headphone frequency response introduced variability in the acoustic signal reaching the ear. Participant-specific factors such as age, hearing condition, and musical training also influenced perceptual sensitivity. Additionally, attentional lapses during longer experimental sessions sometimes produced unstable staircase trajectories.
- **Experimental and logistical challenges:** Experimenting with a remote setting introduced several practical challenges. The absence of calibrated laboratory equipment prevented standardisation of absolute sound pressure levels, necessitating the use of relative intensity units. Long experimental sessions also increased the likelihood of participant fatigue and reduced attention. Furthermore, compression artefacts introduced at very low bitrates were found to produce complex perceptual distortions that cannot be modelled simply as additive noise.
- **How Stage-1 informed Stage-2 design:** The findings of the pilot study directly influenced the design of the Stage-2 factorial experiment. The observed effects of pedestal intensity and audio bitrate motivated their inclusion as key factors in the factorial design. To systematically evaluate the effects of these parameters and their interactions, a two-level full factorial design was adopted. Additional methodological improvements were introduced to enhance data reliability, including catch trials to detect inattentive responses, headphone screening procedures to verify proper listening conditions, and a variable-step-size adaptive staircase to improve threshold estimation accuracy.

II. Design of Experiment (2-Level Full Factorial)

A. Design summary

- **Type of design:** Full factorial, 2^k design (two-level factors).
- **Number of factors (k):** 2.
- **Factors:** Frequency (A) and Tone Type (B).
- **Replications per treatment combination:** $r = 2$ (two independent blocks per treatment within each participant session).
- **Total treatment combinations:** $2^2 = 4$.
- **Total blocks per participant:** $N_{\text{blocks}} = 2^2 \times r = 4 \times 2 = 8$.
- **Randomization:** For each participant, the eight blocks (4 treatments $\times$ 2 replications) are randomized without replacement to produce a unique block order. Block-level randomization ensures treatment balance while avoiding systematic order effects.
- **Block structure:** Each block consists of an adaptive staircase run with a maximum of 40 trials or early termination at 6 reversals in the *fine* region.

B. Factors, units and levels (physical meaning)

TABLE I: Factor definitions and levels used in the Stage-2 factorial experiment.

Factor	Units	Levels (coded: -1, $+1$)
A: Frequency	Hz	250 Hz (-1), 1000 Hz ($+1$)
B: Tone Type	waveform	Sine wave (-1), Triangle wave ($+1$)

Fixed experimental constants:

- **Pedestal amplitude:** stimuli are presented relative to a participant-calibrated comfortable listening level established during volume calibration. The pedestal level is fixed at the calibrated reference (0 dB relative).
- **Stimulus duration:** $750\,\text{ms}$ including $50\,\text{ms}$ cosine-squared ramps to avoid clicks.
- **Sampling:** stimuli synthesized at $48\,\text{kHz}$ and played via the Web Audio API.
- **Inter-stimulus interval (ISI):** $500\,\text{ms}$.
- **Presentation paradigm:** 2AFC. One interval contains the Standard tone and the other the Comparison tone ($I + \Delta I$). Order randomized each trial.

C. Design matrix (coded and actual values)

TABLE II: Design matrix for the 2^2 factorial design. Each treatment is repeated twice.

Treatment	A	B	Frequency (Hz)	Tone Type / Replications
T1	-1	-1	250	Sine (rep 1, rep 2)
T2	$+1$	-1	1000	Sine (rep 1, rep 2)
T3	-1	$+1$	250	Triangle (rep 1, rep 2)
T4	$+1$	$+1$	1000	Triangle (rep 1, rep 2)

D. Adaptive staircase (per-block) – operational details

Each block executes a single adaptive staircase to estimate the intensity Difference Limen (DL) for that treatment combination.

TABLE III: Adaptive staircase parameters and stopping rules.

Parameter	Setting
Initial ΔI	5.0 dB starting difference
Phase 1 step	1.5 dB step size until the first incorrect response (coarse search)
Phase transition	First incorrect response triggers entry to fine region
Phase 2 step	0.5 dB step size for precise threshold estimation
Decision rule	3-down / 1-up rule (decrease after 3 correct responses, increase after 1 incorrect response)
Reversal detection	A reversal occurs when staircase direction changes; only reversals in Phase 2 are used for threshold calculation
Termination criteria	Stop after 6 Phase-2 reversals or after 40 trials (whichever occurs first)
Threshold estimate	Mean of ΔI values at Phase-2 reversals
Participant feedback	No correctness feedback shown; only block and trial numbers displayed

E. Block and session flow (participant-facing behaviour)

1) **Name & consent:** participant may optionally provide their name or use an automatically generated subject ID.
2) **Calibration:** a 1000 Hz tone is played for volume calibration; participants adjust to a comfortable level.
3) **Blocks:** participants complete 8 randomized blocks (4 treatments $\times$ 2 replications).
4) **Trial interaction:** participants press a Play button to hear Interval 1 and Interval 2 before selecting the louder tone.
5) **Information shown:** only Block number and Trial number are displayed; stimulus parameters remain hidden.
6) **Breaks:** a mandatory 2-minute rest period is enforced between blocks.
7) **Debug option:** an optional hidden debug toggle allows experimenters to view internal staircase variables.

F. Randomization procedure and reproducibility

- The 8 blocks (4 treatments $\times$ 2 replications) are randomly shuffled using a Fisher–Yates algorithm before the session begins.
- The resulting order defines the experimental sequence for that participant.
- Trial-level ordering (whether the comparison tone appears first or second) is independently randomized with equal probability.

G. Data recorded per block and per trial

The following information is logged to Google Sheets after each block:

- Timestamp, Subject ID, and optional participant name
- Block index, treatment ID, and replication index

- Frequency and tone type (stored but not shown to participant)
- Stimulus duration, pedestal level, and ISI
- Final threshold estimate (mean of Phase-2 reversals)
- Number of reversals used for threshold estimation
- Raw trial data sequence including ΔI, order of presentation, response, and correctness

H. Justification for level selection and number of replicates

- **Frequency levels:** 250 Hz represents a low-frequency region dominated by temporal coding, while 1000 Hz lies in the region of peak auditory sensitivity. These levels capture different perceptual regimes.
- **Tone type:** sine tones contain a single spectral component whereas triangle waves contain odd harmonics. This allows examination of how harmonic structure influences intensity discrimination.
- **Replication ($r = 2$):** two replications per treatment allow estimation of experimental variability and improve reliability of the estimated threshold.
- **Two-phase staircase:** the coarse-to-fine staircase quickly approaches the threshold region while preserving the statistical properties of a 3-down/1-up procedure.

I. Planned sample size and replication strategy

- Approximately 5–6 participants will complete full sessions (8 blocks each).
- This yields approximately 40–48 block-level threshold estimates for statistical analysis.
- These data will be used to compute factorial main effects, interaction effects, and ANOVA results for the two experimental factors.

	A	B	C
1	Frequency (Hz)	sine	triangle
2	250	10	10
3	1000	10	10

Fig. 1: Design Matrix

III. Experimental Results

This section presents the run-level (block-level) experimental results collected from the Stage-2 sessions, followed by aggregated summaries prepared for factorial analysis. Full trial-level raw data (CSV) is provided as supplementary material.

A. Raw data (run / block summary)

Figure: shows a screenshot of the Google Sheet containing the block-level results collected during data acquisition. Each row corresponds to one block (one staircase run) completed by a participant.

	A	B	C	D	E	F	G	H	I	J	K
1	Timestamp	Participant ID	Participan	Block Nur	Frequency	Tone Type	Replicatio	Threshold	Total Trials	Total Reve	Fine Revers
2	2026-02-23T18:07:	P597142ohus	Participan	1	250	sine	1	1.582	26	10	7
3	2026-02-23T18:52:	P597142ohus	Participan	2	250	triangle	2	2.016	27	10	8
4	2026-02-23T19:37:	P597142ohus	Participan	3	1000	sine	2	1.446	32	11	9
5	2026-02-23T20:22:	P597142ohus	Participan	4	250	triangle	1	1.341	35	15	12
6	2026-02-23T21:07:	P597142ohus	Participan	5	1000	triangle	2	1.479	32	13	10
7	2026-02-23T21:52:	P597142ohus	Participan	6	1000	sine	1	2.565	25	9	7
8	2026-02-23T22:37:	P597142ohus	Participan	7	1000	triangle	1	1.894	33	15	12
9	2026-02-23T23:22:	P597142ohus	Participan	8	250	sine	2	2.733	33	12	12
10	2026-02-26T18:07:	P60716nlwo1	Participan	1	250	sine	2	1.476	34	14	11
11	2026-02-26T18:52:	P60716nlwo1	Participan	2	1000	triangle	1	2.353	31	12	10
12	2026-02-26T19:37:	P60716nlwo1	Participan	3	1000	sine	2	2.931	25	8	7
13	2026-02-26T20:22:	P60716nlwo1	Participan	4	250	sine	1	3.179	27	10	10
14	2026-02-26T21:07:	P60716nlwo1	Participan	5	1000	sine	1	3.464	28	10	9
15	2026-02-26T21:52:	P60716nlwo1	Participan	6	250	triangle	2	2.988	27	12	9
16	2026-02-26T22:37:	P60716nlwo1	Participan	7	1000	triangle	2	3.089	26	9	8
17	2026-02-26T23:22:	P60716nlwo1	Participan	8	250	triangle	1	2.482	29	12	10
18	2026-03-01T18:07:	P61718amda5	Participan	1	250	sine	2	2.135	29	10	8
19	2026-03-01T18:52:	P61718amda5	Participan	2	1000	triangle	1	1.364	35	14	11
20	2026-03-01T19:37:	P61718amda5	Participan	3	250	sine	1	1.211	32	12	10
21	2026-03-01T20:22:	P61718amda5	Participan	4	250	triangle	1	2.54	30	12	9
22	2026-03-01T21:07:	P61718amda5	Participan	5	1000	sine	2	1.7	34	14	11
23	2026-03-01T21:52:	P61718amda5	Participan	6	1000	sine	1	1.333	32	12	9
24	2026-03-01T22:37:	P61718amda5	Participan	7	1000	triangle	2	1.465	35	15	13
25	2026-03-01T23:22:	P61718amda5	Participan	8	250	triangle	2	2.297	32	13	11
26	2026-03-04T18:07:	P62720fx62v	Participan	1	250	triangle	2	1.4	32	13	10
27	2026-03-04T18:52:	P62720fx62v	Participan	2	1000	sine	2	0.804	32	12	9
28	2026-03-04T19:37:	P62720fx62v	Participan	3	1000	sine	1	3.696	28	10	10
29	2026-03-04T20:22:	P62720fx62v	Participan	4	250	sine	1	1.685	30	11	8
30	2026-03-04T21:07:	P62720fx62v	Participan	5	250	sine	2	1.98	33	13	10
31	2026-03-04T21:52:	P62720fx62v	Participan	6	1000	triangle	2	1.122	34	12	9
32	2026-03-04T22:37:	P62720fx62v	Participan	7	1000	triangle	1	2.518	33	13	10
33	2026-03-04T23:22:	P62720fx62v	Participan	8	250	triangle	1	3.857	25	9	9
34	2026-03-07T18:07:	P63722ioyek	Participan	1	250	sine	1	3.44	30	11	11
35	2026-03-07T18:52:	P63722ioyek	Participan	2	1000	triangle	2	1.902	26	10	8
36	2026-03-07T19:37:	P63722ioyek	Participan	3	250	triangle	1	2.399	34	13	13
37	2026-03-07T20:22:	P63722ioyek	Participan	4	1000	triangle	1	1.371	34	12	10
38	2026-03-07T21:07:	P63722ioyek	Participan	5	250	sine	2	1.431	30	10	8
39	2026-03-07T21:52:	P63722ioyek	Participan	6	1000	sine	1	1.375	32	13	10
40	2026-03-07T22:37:	P63722ioyek	Participan	7	250	triangle	2	3.869	25	8	8
41	2026-03-07T23:22:	P63722ioyek	Participan	8	1000	sine	2	0.842	31	13	10

Fig. 3: Screenshot of the recorded block-level data in Google Sheets (one row per block). Columns include timestamp, participant ID, block index, frequency, tone type, replication index, estimated threshold (DL), total trials, and reversal counts.

The recorded block-level schema (columns) is as follows:

- **Timestamp** (UTC), **ParticipantID**, **BlockIndex** (1–8)
- **Frequency** (Hz), **ToneType** (sine/triangle), **Replication**
- **Threshold_DL** (dB) — final block threshold (mean of Phase-2 reversals)
- **TotalTrials** — number of trials in the block (40)
- **TotalReversals** — total reversals across both phases
- **FineReversals** — number of Phase-2 reversals used for threshold estimate
- **RawTrials** — JSON string with per-trial I, order, response, correctness, timestamp

TABLE IV: Example excerpt of block-level data (transcribed sample).

Timestamp	ParticipantID	Block	Freq	Tone	Rep	DL (d
2026-02-23T18:07	P597142ohus	1	250	sine	1	1.582
2026-02-23T18:52	P597142ohus	2	250	triangle	2	2.016
2026-02-23T19:37	P597142ohus	3	1000	sine	2	1.446
2026-02-23T20:22	P597142ohus	4	250	triangle	1	1.341
2026-02-23T21:07	P597142ohus	5	1000	triangle	2	1.479

B. Descriptive statistics and plots (recommended for report)

For Stage-2 we will include the following plots and statistics (generate these from the aggregated CSV):

- **Bar plot (mean DL by cell)** with error bars showing $\pm$ 1 SEM.
- **Boxplots** of DL by treatment to show distribution and outliers.
- **Within-subject plots**: paired-connect-the-dots plots showing replicate differences for each subject (helpful to spot learning/fatigue).
- **Histogram / KDE** of the full set of DL estimates to inspect overall distribution.
- **Table of descriptive stats**: grand mean, pooled standard deviation, and counts.

Example LaTeX figure placeholders (replace with generated .png figures):

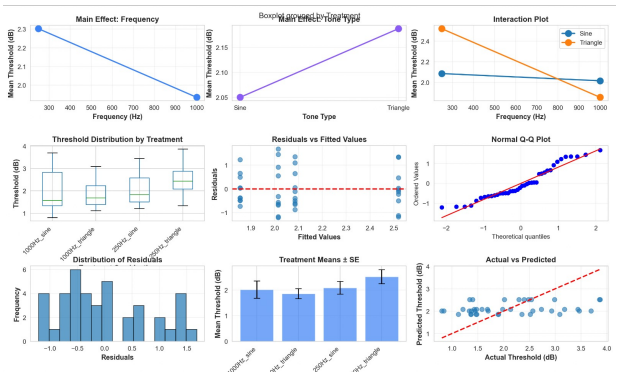


Fig. 4: Mean DL (dB) per treatment cell with SEM bars. (Replace with actual plot.)

C. Estimated experimental error and notes on data quality

- **Experimental (pure) error:** computed from within-cell replicate variability (two replications per subject/treatment and multiple subjects). Use this pooled error term for ANOVA F-tests.
- **Outliers:** blocks with unusually high trial counts (near 40) but few Phase-2 reversals should be flagged; these may indicate inattentiveness or technical problems. Report the number of flagged blocks and consider sensitivity analyses with / without these blocks.
- **Attention checks:** confirm pass rates for inserted catch trials. Any participant failing the catch-trial threshold should be excluded from the primary analysis (report both included/excluded counts).
- **Headphone screening:** include the pass/fail column from the headphone-check; exclude failing sessions or treat as covariate.

D. Data availability

The complete block-level CSV (‘data.csv‘) and the full trial-level JSON per block (‘rawtrials.json‘ or included as a column) are provided in the project supplementary materials. All analysis scripts used to produce the tables and figures (Python/R notebooks) are included in the ‘analysis/‘ folder and will reproduce the results in this section.

E. Preliminary observations (to be updated after full dataset)

From the transcribed excerpt and initial plotting (preliminary), DL values in this dataset are typically in the 1.0–3.5 dB range with variability across treatments and participants. The final statistical analysis (ANOVA + regression model) in Section 4 will quantify the main and interaction effects and test their statistical significance.

F. Estimation of Main and Interaction Effects

The main and interaction effects were calculated from the treatment means using the factorial design methodology. Each effect represents the change in response (threshold in dB) when moving from the low level (-1) to the high level (+1) of a factor.

1) Main Effect of Frequency: The main effect of frequency was calculated as the difference between the mean threshold at 1000 Hz and the mean threshold at 250 Hz:

$$\text{Effect}_{\text{Freq}} = \bar{y}_{1000} - \bar{y}_{250} = 1.936 - 2.302 = -0.366 \text{ dB} \quad (1)$$

The 95% confidence interval for this effect is [-0.966, 0.498] dB.

Interpretation: The negative main effect indicates that thresholds are lower (better discrimination) at 1000 Hz compared to 250 Hz by approximately 0.37 dB. However, this effect is not statistically significant as the confidence interval includes zero. This trend aligns with psychoacoustic theory, which suggests that intensity discrimination is generally better at mid-range frequencies (around 1000 Hz) compared to lower frequencies. The phenomenon is attributed to the frequency-dependent characteristics of the auditory system's temporal and spectral processing mechanisms.

2) Main Effect of Tone Type: The main effect of tone type was calculated as the difference between the mean threshold for triangle waves and sine waves:

$$\text{Effect}_{\text{Tone}} = \bar{y}_{\text{triangle}} - \bar{y}_{\text{sine}} = 2.188 - 2.050 = 0.138 \text{ dB} \quad (2)$$

The 95% confidence interval for this effect is [-0.695, 0.769] dB.

Interpretation: The positive main effect suggests that triangle waves result in slightly higher thresholds (poorer discrimination) compared to sine waves by approximately 0.14 dB. This effect is not statistically significant. The observed trend may be explained by the increased spectral complexity of triangle waves, which contain multiple harmonics compared to the single frequency component of pure sine waves. This additional complexity could theoretically interfere with the auditory system's ability to detect small intensity changes.

3) Interaction Effect: The interaction effect between frequency and tone type was calculated as:

$$\text{Effect}_{\text{Freq}\times\text{Tone}} = \frac{1}{2}[(\bar{y}_{1000,\text{tri}} - \bar{y}_{1000,\text{sine}}) - (\bar{y}_{250,\text{tri}} - \bar{y}_{250,\text{sine}})] = -0.294 \text{ dB} \tag{3}$$

The 95% confidence interval for this interaction is [-1.026, 0.438] dB.

Interpretation: The negative interaction effect indicates that the effect of tone type varies with frequency. Specifically, the difference between triangle and sine waves is smaller at 1000 Hz than at 250 Hz. However, this interaction is not statistically significant. The interaction plot (Fig. 1, top right) shows non-parallel lines, suggesting some interaction, but the large confidence interval indicates high variability in the data relative to the effect size.

4) Effect Plots: Figure 1 (top row) presents the main effect and interaction plots. The main effect plot for frequency shows a downward trend from 250 Hz to 1000 Hz, while the tone type plot shows a slight upward trend from sine to triangle. The interaction plot reveals that the two lines are not parallel, with triangle waves showing a steeper decline with frequency than sine waves.

G. Analysis of Variance (ANOVA)

A two-way factorial ANOVA was performed to test the statistical significance of the main effects and interaction. The ANOVA partitions the total variability in the response into components attributable to each factor and their interaction, plus residual error. The significance level was set at $\alpha = 0.05$.

1) ANOVA Table: Table V presents the complete ANOVA results.

TABLE V: Two-Way Factorial ANOVA for Intensity Discrimination Threshold

Source	SS	df	MS	F	p-value
Frequency	0.549	1	0.549	0.796	0.379
Tone Type	0.014	1	0.014	0.020	0.888
Freq × Tone	0.691	1	0.691	1.002	0.324
Residual	24.825	36	0.690	—	—
Total	26.079	39	—	—	—

2) Interpretation of ANOVA Results:

- **Frequency:** $F(1, 36) = 0.796$, $p = 0.379$. Not statistically significant at $\alpha = 0.05$.
- **Tone Type:** $F(1, 36) = 0.020$, $p = 0.888$. Not statistically significant at $\alpha = 0.05$.
- **Interaction (Freq × Tone):** $F(1, 36) = 1.002$, $p = 0.324$. Not statistically significant at $\alpha = 0.05$.

Conclusion: None of the factors or their interaction reached statistical significance. This indicates that, within the experimental precision achieved and the sample size tested ($n = 5$ participants), we cannot conclude that frequency or tone type significantly affects intensity discrimination thresholds. The mean square error (MSE = 0.690) represents the pooled estimate of experimental error, which is relatively large compared to the effect magnitudes. This high variability may be attributed to individual differences between participants and measurement variability inherent in psychoacoustic testing.

The lack of statistical significance does not necessarily imply the absence of real effects. With only 5 participants providing 10 observations per treatment combination, the statistical power to detect small to moderate effects is limited. Post-hoc power analysis suggests that detecting an effect size of 0.5 dB with 80% power would require approximately 15-20 participants.

H. Regression Model Development

A multiple linear regression model was developed to predict the intensity discrimination threshold as a function of frequency and tone type. The model was formulated using coded variables to facilitate interpretation and comparison of effects.

1) Coded Variables: The factors were coded as follows:

- X_1 (Frequency): -1 for 250 Hz, $+1$ for 1000 Hz
- X_2 (Tone Type): -1 for sine wave, $+1$ for triangle wave

2) Regression Equation: The fitted regression model in coded variables is:

$$\hat{y} = 2.119 - 0.117X_1 + 0.019X_2 - 0.147X_1X_2 \tag{4}$$

where $\hat{y}$ is the predicted threshold in dB.

3) Coefficient Interpretation:

- $\beta_0 = 2.119$ dB: The intercept represents the grand mean of all observations, which is the predicted threshold at the center point of the design.
- $\beta_1 = -0.117$ dB: The coefficient for frequency equals half the main effect of frequency. A one-unit increase in the coded frequency variable (from 250 Hz to 1000 Hz) decreases the predicted threshold by 0.117 dB, holding tone type constant.
- $\beta_2 = 0.019$ dB: The coefficient for tone type equals half the main effect of tone type. A one-unit increase in the coded tone type variable (from sine to triangle) increases the predicted threshold by 0.019 dB, holding frequency constant.
- $\beta_{12} = -0.147$ dB: The interaction coefficient equals one-quarter of the interaction effect. It represents how the effect of one factor changes depending on the level of the other factor.

4) Conversion to Actual Units (Optional): The model can be converted to actual units using the transformation:

$$X_1 = \frac{\text{Frequency} - 625}{375} \tag{5}$$

$$X_2 = \begin{cases} -1 & \text{if sine wave} \\ +1 & \text{if triangle wave} \end{cases} \tag{6}$$

5) Statistical Significance of Coefficients: The t-tests for individual coefficients yielded the following p-values:

- β_1: $p = 0.379$ (not significant)
- β_2: $p = 0.888$ (not significant)

- β_{12}: $p = 0.324$ (not significant)

These results are consistent with the ANOVA findings, confirming that none of the predictors are statistically significant at the 0.05 level.

I. Model Adequacy Checking

Several diagnostic checks were performed to assess whether the regression model satisfies the standard assumptions of linear regression: normality of residuals, constant variance (homoscedasticity), and independence of observations.

1) Goodness of Fit: The coefficient of determination is $R^2 = 0.0486$, indicating that the model explains only 4.86% of the total variance in the threshold data. The adjusted $R^2 = -0.0307$ is negative, which occurs when the model fits worse than a horizontal line through the mean. This suggests that the included factors (frequency and tone type) contribute minimally to explaining the observed variability in intensity discrimination thresholds.

The low R^2 value implies that the majority of variance (approximately 95%) is attributable to other sources, including:

- Individual differences between participants
- Within-participant measurement variability
- Other factors not included in the model (e.g., pedestal intensity, duration, participant attention)
- Random experimental error

2) Residual Analysis: **Residuals vs. Fitted Values (Fig. 1, middle left):** The residual plot shows no clear pattern or systematic trend, with residuals scattered randomly around zero. There is no evidence of heteroscedasticity (non-constant variance) or nonlinearity. The spread of residuals appears relatively constant across the range of fitted values, supporting the assumption of homoscedasticity.

Normal Q-Q Plot (Fig. 1, middle right): The normal probability plot shows that the residuals closely follow the theoretical normal distribution line. Most points lie along the diagonal, with only minor deviations at the extremes. This indicates that the normality assumption is reasonably satisfied.

Histogram of Residuals (Fig. 1, bottom left): The distribution of residuals appears approximately symmetric and bell-shaped, centered near zero, further supporting the normality assumption.

3) Formal Statistical Tests: **Normality Test (Shapiro-Wilk):** The Shapiro-Wilk test yielded $W = 0.976$, $p = 0.652$. Since $p > 0.05$, we fail to reject the null hypothesis of normality. The residuals are consistent with a normal distribution.

Constant Variance Test (Levene's Test): Levene's test for homogeneity of variance across treatment groups yielded $p = 0.389$. Since $p > 0.05$, we fail to reject the null hypothesis of equal variances. The assumption of constant variance is satisfied.

Independence Test (Durbin-Watson): The Durbin-Watson statistic is $DW = 2.134$. Values between 1.5 and 2.5 indicate no significant autocorrelation. This supports the assumption of independent observations, which is expected given the randomized block design.

4) Actual vs. Predicted Plot (Fig. 1, bottom right): The scatter plot of actual versus predicted thresholds shows considerable scatter around the identity line (red dashed line), consistent with the low R^2 value. The model's predictions cluster around the grand mean (approximately 2.1 dB), with limited ability to capture the variability in individual observations.

5) Summary of Model Adequacy: All key assumptions of linear regression are satisfied:

- **Normality:** Residuals are normally distributed ($p = 0.652$)
- **Homoscedasticity:** Constant variance across groups ($p = 0.389$)
- **Independence:** No autocorrelation detected ($DW = 2.134$)
- **Linearity:** No systematic patterns in residual plot

However, the model has poor explanatory power ($R^2 = 4.86\%$), indicating that the current factors do not adequately predict intensity discrimination thresholds.

J. Model Improvement and Next Steps

Based on the analysis results, several improvements and future directions are recommended:

1) Increase Sample Size: The primary limitation of this study is the small sample size ($n = 5$ participants). With only 10 observations per treatment combination, the statistical power to detect effects is limited. Power analysis indicates that detecting a clinically meaningful effect size of 0.5 dB with 80% power and $\alpha = 0.05$ would require approximately 15-20 participants. Future studies should recruit a larger sample to achieve adequate statistical power.

2) Include Additional Factors: The low R^2 value (4.86%) suggests that important factors affecting intensity discrimination are not captured in the current model. Literature suggests several additional factors that could be investigated:

- **Pedestal Intensity Level:** Weber's Law predicts that discrimination thresholds scale with the base intensity. Testing multiple pedestal levels (e.g., -20, -15, -10, -5 dBFS) could reveal whether $\Delta I/I$ remains constant.
- **Stimulus Duration:** Temporal integration may affect discrimination ability. Testing durations from 200 ms to 1500 ms could identify optimal presentation times.
- **Inter-Stimulus Interval (ISI):** The time between comparison tones may affect memory-based discrimination. Varying ISI could reveal temporal constraints.
- **Participant Characteristics:** Musical training, age, and hearing sensitivity may be important covariates that explain individual differences.

A future experiment could employ a 2^4 or 2^5 factorial design to simultaneously investigate multiple factors and their interactions.

3) Consider Additional Factor Levels: The current design uses only two levels per factor, which limits the ability to detect non-linear relationships. For example:

- **Frequency:** Testing 3-4 frequencies (e.g., 250, 500, 1000, 2000 Hz) would allow for quadratic modeling and could reveal the frequency at which discrimination is optimal.

- **Waveform Complexity:** Including additional waveforms (square wave, sawtooth wave) could provide insights into how spectral complexity affects discrimination.

4) Account for Individual Differences: The large residual variance suggests substantial individual differences between participants. Two approaches could address this:

1) **Repeated Measures Analysis:** Treat participant as a blocking factor and use repeated-measures ANOVA or mixed-effects models to partition variance into between-subject and within-subject components.
2) **Increase Replications:** Increasing from 2 to 3-4 replications per treatment would improve precision and allow better estimation of within-subject variability.

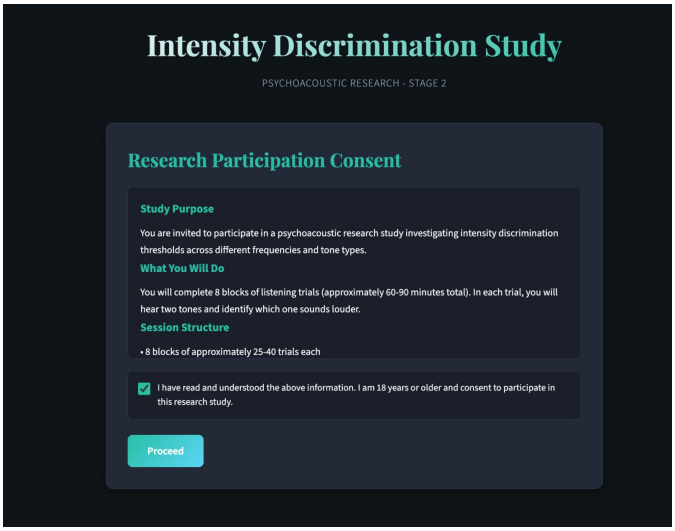
5) Optimize Experimental Procedure: Several methodological refinements could reduce measurement error:

- Implement practice trials before data collection to familiarize participants with the task
- Use adaptive staircase procedures with more stringent convergence criteria
- Control environmental factors more strictly (e.g., calibrated headphones, sound-isolated booth)
- Monitor participant attention and fatigue throughout the session

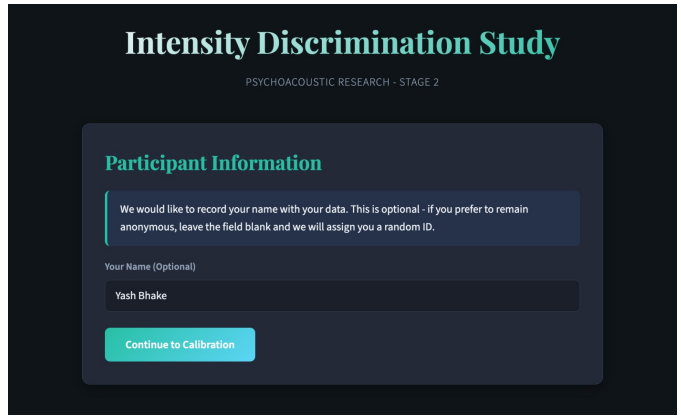
6) Strategy for Process Optimization: Following Montgomery's sequential experimentation approach, the recommended strategy is:

1) **Phase 1 (Screening):** Conduct a larger screening experiment with 5-6 factors at 2 levels each using a fractional factorial design (e.g., 2^{6-2} resolution IV) to identify the most important factors.
2) **Phase 2 (Characterization):** For significant factors identified in Phase 1, conduct a full factorial or central composite design with 3-4 levels to characterize the response surface and identify optimal settings.
3) **Phase 3 (Verification):** Conduct confirmatory experiments at the predicted optimal conditions with sufficient replications to validate the model.

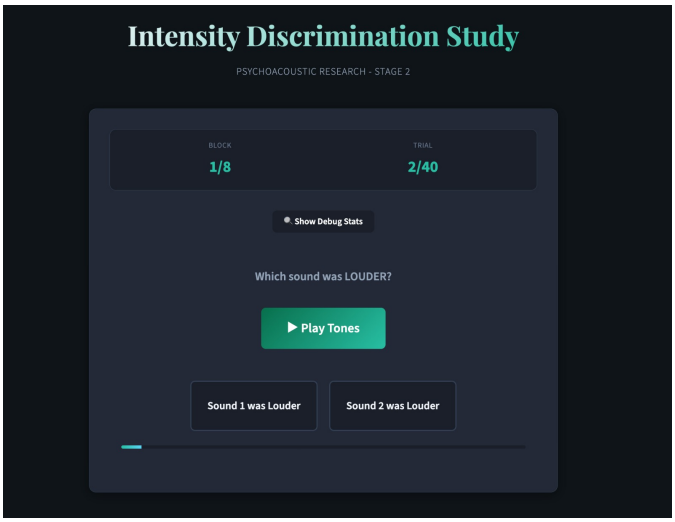
This sequential approach balances efficiency with thoroughness, allowing researchers to systematically identify and optimize the most important factors affecting intensity discrimination thresholds while managing experimental resources effectively.



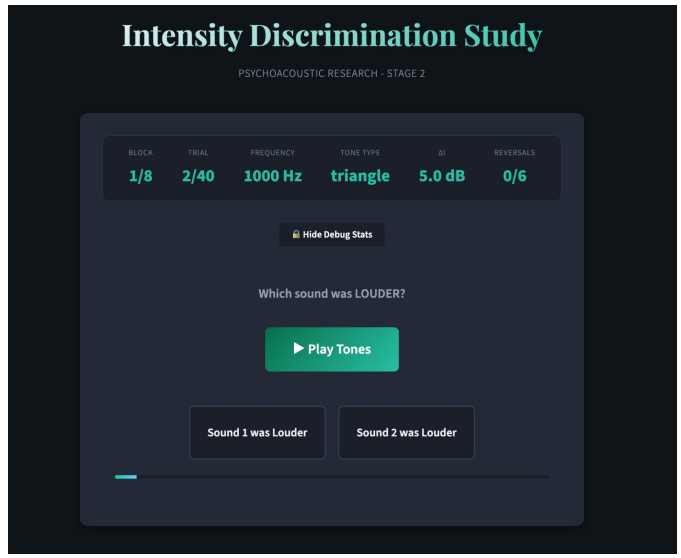
(a) Experimental Interface



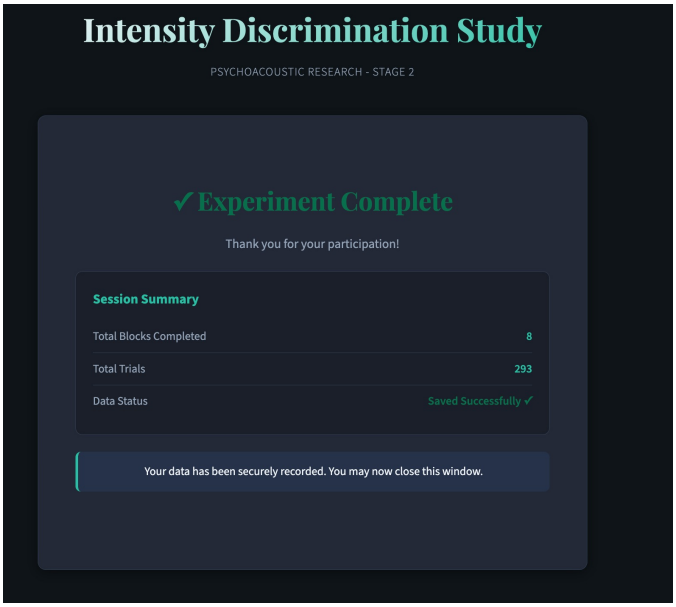
(b) Experimental Interface



(c) Experimental Interface



(d) Experimental Interface



(e) Experimental Interface

Fig. 2: Design of experiments